

Task 2

(1)

```
## Call:
## ergm(formula = glasgow_net1 ~ edges + nodematch("gender"))
##
## Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -5.0133    0.1365      0 -36.72  <1e-04 ***
## nodematch.gender  2.0098    0.1460      0  13.77  <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 3824 on 16510 degrees of freedom
##
## AIC: 3828 BIC: 3843 (Smaller is better. MC Std. Err. = 0)
```

- Negative (and significant) edge coefficient - networks tend to be sparse
- Positive (and significant) homophily coefficient - networks tend to have homophily

```
## Conditional probability of a tie between same-gender students: 0.04726488
```

(2)

```
ergm_model_2_params <- glasgow_net1 ~
  # Model 1
  edges +
  nodematch("gender") +

  # Endogenous Hypotheses
  mutual +
  gwesp(0.2, fixed = TRUE) +

  # Exogenous Hypotheses: Gender
  nodeofactor("gender", levels = 1) +
  nodeifactor("gender", levels = 1) +

  # Exogenous Hypotheses: Alcohol
  nodecov("alcohol") +
  nodeicov("alcohol") +
  absdiff("alcohol") +
```

```

# Basic Exogenous Controls
edgecov(logdistance_matrix) +
absdiff("age") +

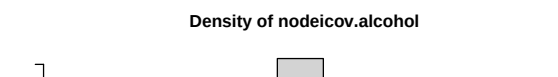
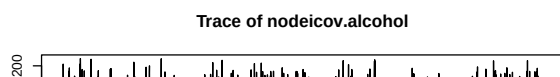
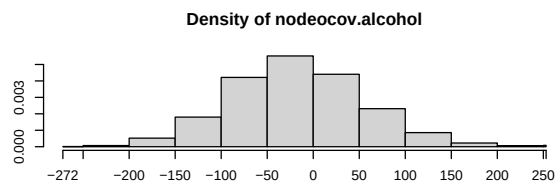
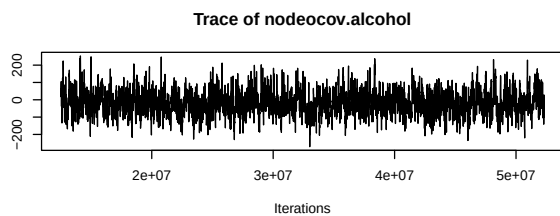
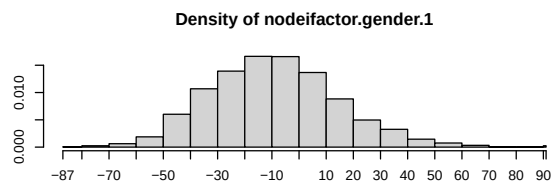
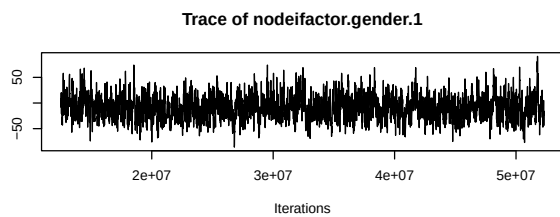
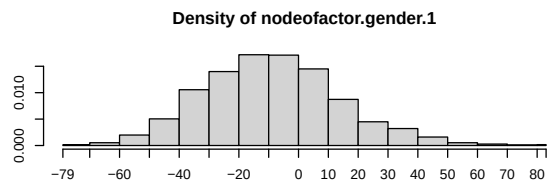
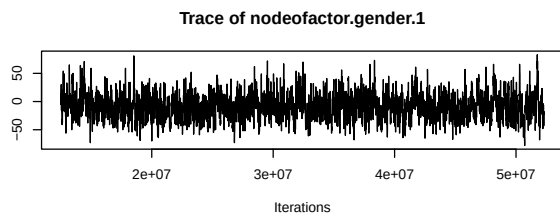
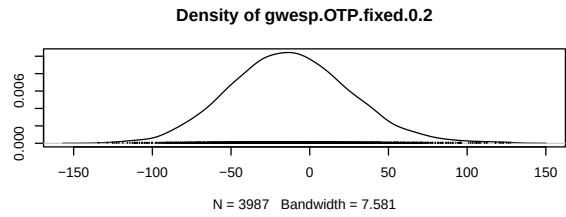
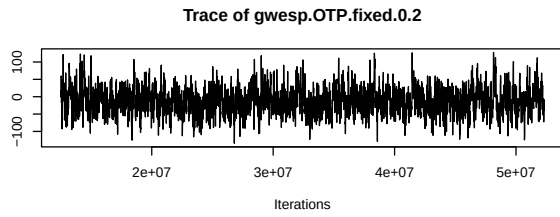
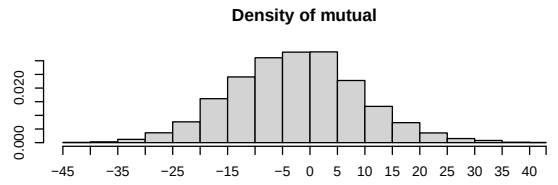
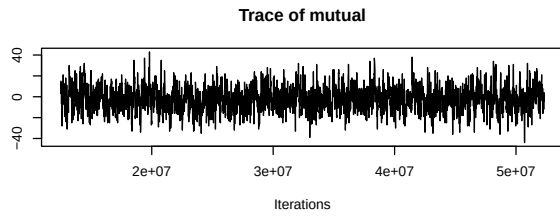
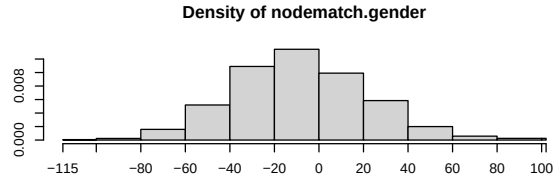
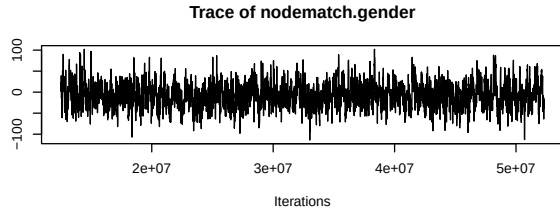
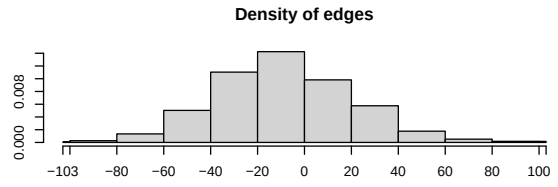
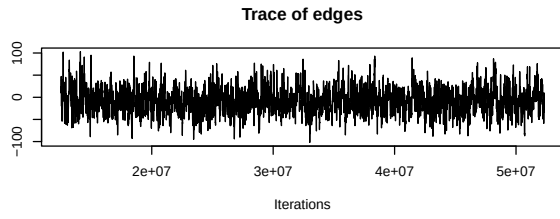
# Basic Endogenous Controls (Stabilizers)
gwidegree(0.6, fixed = TRUE) +
gwodegree(0.6, fixed = TRUE)
noisy_model_2 <- ergm(
  ergm_model_2_params,
  # ERGMs with this many parameters can take a moment to compute
  control = control.ergm(
    MCMC.samplesize = 5000,
    MCMC.interval = 8000,
    MCMC.burnin = 1 * 1000 * 1000,
    main.method = "Stochastic-Approximation",

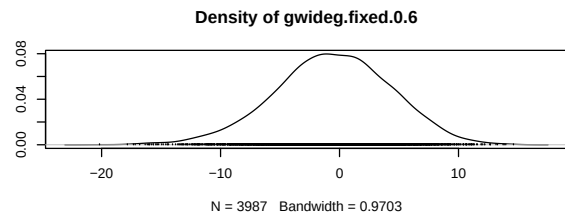
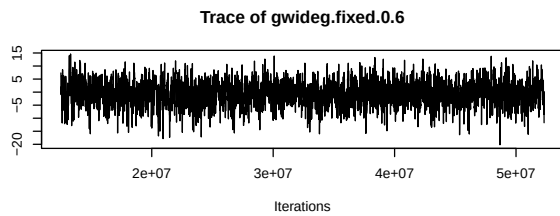
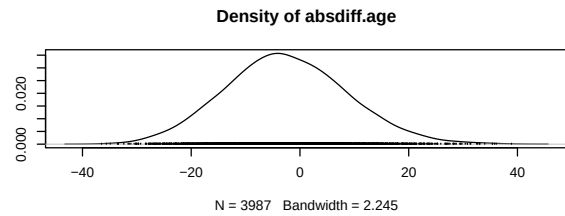
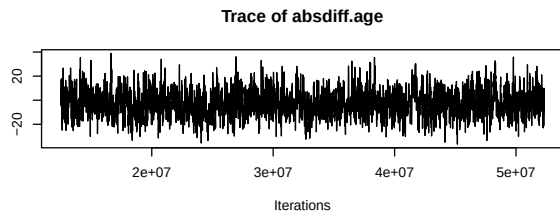
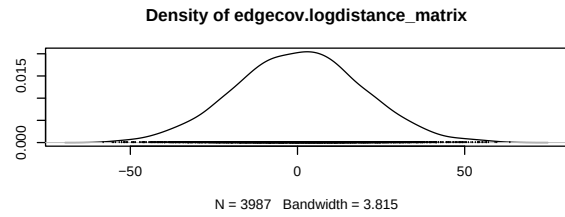
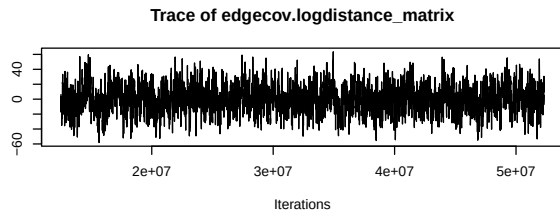
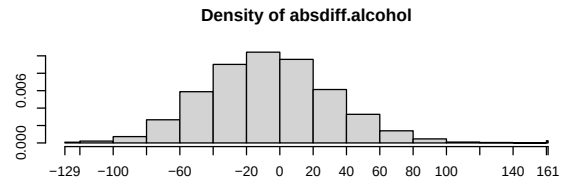
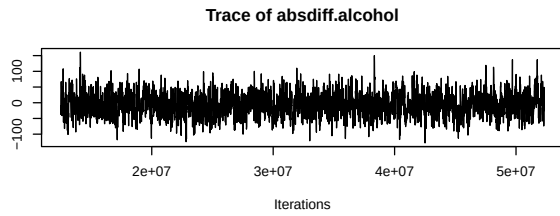
    parallel = 8,
    parallel.type = "PSOCK"
  )
)
rough_estimates <- coef(noisy_model_2)
model_2 <- ergm(
  ergm_model_2_params,
  control = control.ergm(
    MCMC.samplesize = 5000,
    MCMC.interval = 10 * 1000,
    MCMC.burnin = 100 * 1000,
    init = rough_estimates,

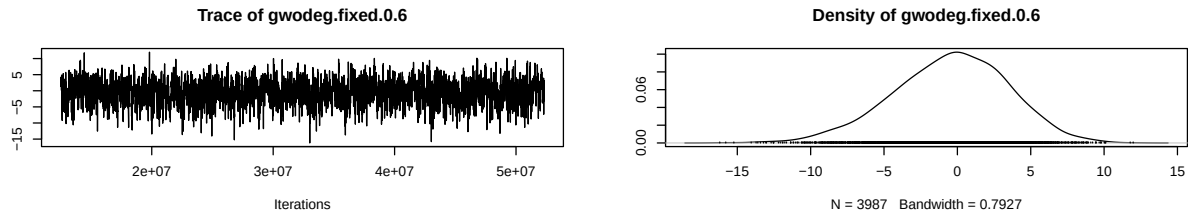
    parallel = 8,
    parallel.type = "PSOCK"
  )
)

```

(3)







We see that the distributions are unimodal and mostly centered around 0. The traces oscillate around 0 without any patterns.

```
## , , edges
##
##          edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0      1.00000000      0.94312183 0.84340578      0.94607114
## Lag 9984    0.57259725      0.58542783 0.55875965      0.58711266
## Lag 49920   0.16306081      0.17397730 0.15530825      0.16969929
## Lag 99840   0.02397634      0.02490695 0.02959222      0.03245918
## Lag 499200 0.01954303      0.03328058 0.01718032      0.01917022
##          nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0      0.80058239      0.77975242      0.94394962
## Lag 9984    0.46588728      0.45811691      0.53648962
## Lag 49920   0.12753813      0.12952367      0.14167521
## Lag 99840   0.02074313      0.01391207      0.01723353
## Lag 499200 0.02050540      0.02501228      0.01060647
##          nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0      0.930848976      0.692827732      0.16102644
## Lag 9984    0.526240353      0.357997066      0.01707464
## Lag 49920   0.132779861      0.053911314      -0.04182880
## Lag 99840   0.015422576     -0.031773736      -0.04118444
```

```

## Lag 499200      0.005377521      0.006964754      -0.01530326
##      absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0      0.80762722      0.680387792      0.7605902279
## Lag 9984      0.44294133      0.343143323      0.3804058948
## Lag 49920      0.12498785      0.083446867      0.0968683379
## Lag 99840      0.01252297      0.023015281      0.0001874916
## Lag 499200      0.02562230      -0.006764463      -0.0036984599
##
## , , nodematch.gender
##
##      edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0      0.94312183      1.0000000 0.83363013      0.92263741
## Lag 9984      0.58441842      0.6256229 0.57349958      0.60407758
## Lag 49920      0.16655605      0.1816669 0.16258434      0.17315972
## Lag 99840      0.01414309      0.0167516 0.02395097      0.02464583
## Lag 499200      0.01500189      0.0258200 0.01310143      0.01180972
##      nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0      0.75276833      0.733502756      0.890193006
## Lag 9984      0.46290378      0.456509254      0.550230747
## Lag 49920      0.12909477      0.128359318      0.146540214
## Lag 99840      0.01269416      0.006508455      0.008515433
## Lag 499200      0.02375544      0.028302923      0.006189457
##      nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0      0.874205566      0.644812906      0.14775809
## Lag 9984      0.536243169      0.368971906      0.02335643
## Lag 49920      0.136148185      0.056194843      -0.04580299
## Lag 99840      0.006417184      -0.034621620      -0.04319739
## Lag 499200      0.002161112      0.009864441      -0.02076335
##      absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0      0.759634994      0.63755999      0.713055328
## Lag 9984      0.450191264      0.35957993      0.398848213
## Lag 49920      0.127054195      0.09324329      0.099412794
## Lag 99840      0.006090058      0.01996748      -0.002814246
## Lag 499200      0.023692421      -0.01216911      -0.007333763
##
## , , mutual
##
##      edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0      0.84340578      0.83363013 1.000000000      0.893272199
## Lag 9984      0.57738297      0.59465467 0.613104579      0.608201586
## Lag 49920      0.15506133      0.16910690 0.155115775      0.166342829
## Lag 99840      0.01603004      0.01605931 0.026176372      0.027164705
## Lag 499200      0.01123678      0.02322667 0.008605612      0.009852247
##      nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0      0.667645453      0.6534514384      0.79970365
## Lag 9984      0.456879883      0.4479578636      0.54077956
## Lag 49920      0.117979563      0.1212163642      0.13138012
## Lag 99840      0.004044888      -0.0001206603      0.01357055
## Lag 499200      0.019712135      0.0255558713      0.00464238
##      nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0      0.765672236      0.574603932      0.13435411
## Lag 9984      0.525051081      0.373482242      0.04737393
## Lag 49920      0.118877698      0.051895080      -0.04478000
## Lag 99840      0.012220208      -0.028645643      -0.05664441

```

```

## Lag 499200      -0.001347319      0.002617319      -0.01850174
##               absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0          0.675429949      0.63966316      6.799288e-01
## Lag 9984       0.453909220      0.42000171      4.451752e-01
## Lag 49920      0.119218071      0.09703455      9.821924e-02
## Lag 99840      0.006310881      0.01215567      9.856148e-05
## Lag 499200     0.028756831      -0.01586715      -7.111159e-03
##
## , , gwesp.OTP.fixed.0.2
##
##               edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0          0.94607114      0.92263741 0.893272199      1.000000000
## Lag 9984       0.58582059      0.60405158 0.584873310      0.609432217
## Lag 49920      0.16754061      0.18065759 0.165334870      0.177291940
## Lag 99840      0.03155055      0.03373188 0.039459505      0.041561235
## Lag 499200     0.01136560      0.02214827 0.007672245      0.009719613
##               nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0          0.75968828      0.74102190      0.893538085
## Lag 9984       0.47274214      0.46436145      0.550025052
## Lag 49920      0.12854709      0.12833616      0.145068121
## Lag 99840      0.01979188      0.01339045      0.027770278
## Lag 499200     0.01280592      0.01894541      0.004317517
##               nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0          0.874922506      0.645812782      0.13185077
## Lag 9984       0.537871342      0.370326660      0.02374198
## Lag 49920      0.137756466      0.057847843      -0.04252222
## Lag 99840      0.024325763      -0.025613574      -0.05071959
## Lag 499200     -0.000597082      0.003376633      -0.02156532
##               absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0          0.75982364      0.61301936      0.695684130
## Lag 9984       0.45481423      0.36862974      0.404820378
## Lag 49920      0.12917565      0.08721488      0.100571001
## Lag 99840      0.02007183      0.02870936      0.005893387
## Lag 499200     0.02238718      -0.01400073      -0.010157388
##
## , , nodeofactor.gender.1
##
##               edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0          0.80058239      0.75276833 0.66764545      0.75968828
## Lag 9984       0.46274535      0.46360742 0.44122928      0.47189400
## Lag 49920      0.13225196      0.13828228 0.12549738      0.13815706
## Lag 99840      0.04176778      0.03834837 0.04261291      0.04312365
## Lag 499200     0.01881478      0.02750832 0.01681292      0.01280778
##               nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0          1.00000000      0.95651876      0.68130419
## Lag 9984       0.57167999      0.57552845      0.39130900
## Lag 49920      0.15576846      0.16181421      0.10300934
## Lag 99840      0.03718965      0.03248693      0.03389530
## Lag 499200     0.01145838      0.01478887      0.01335874
##               nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0          0.67471626      0.50159226      0.12721293
## Lag 9984       0.38187585      0.25986922      0.01957333
## Lag 49920      0.09312425      0.02895360      -0.04268739
## Lag 99840      0.02956897      -0.03115816      -0.03263282

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## Lag 499200      0.01145863      0.02234388      -0.03175287
##      absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0      0.65372540      0.510070246      0.558760998
## Lag 9984      0.36021987      0.265463986      0.280408171
## Lag 49920      0.09125617      0.064186051      0.069997432
## Lag 99840      0.02748973      0.032150019      0.017066040
## Lag 499200      0.01695467      0.001949188      0.002856944
##
## , , nodeifactor.gender.1
##
##      edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0      0.77975242      0.73350276 0.65345144      0.74102190
## Lag 9984      0.45321369      0.45431523 0.43218124      0.46094003
## Lag 49920      0.13960462      0.14391466 0.13032242      0.14415824
## Lag 99840      0.04081206      0.03955126 0.03887740      0.04040665
## Lag 499200      0.01811821      0.02853621 0.01469033      0.01056136
##      nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0      0.95651876      1.00000000      0.66364488
## Lag 9984      0.57141082      0.57812249      0.38024788
## Lag 49920      0.15904324      0.16778827      0.11184791
## Lag 99840      0.03601767      0.03221758      0.03541748
## Lag 499200      0.01091901      0.01501227      0.01314031
##      nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0      0.64290097      0.47788680      0.12573955
## Lag 9984      0.37039803      0.25006116      0.02134103
## Lag 49920      0.10253567      0.04054971      -0.03359014
## Lag 99840      0.03265806      -0.02721836      -0.03098684
## Lag 499200      0.01001463      0.02278499      -0.03068624
##      absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0      0.63734963      0.492971555      0.543141655
## Lag 9984      0.34932822      0.256510270      0.274467931
## Lag 49920      0.10082859      0.073610089      0.076167498
## Lag 99840      0.02479883      0.029403028      0.016377210
## Lag 499200      0.01632153      0.005417994      0.002739152
##
## , , nodeocov.alcohol
##
##      edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0      0.94394962      0.89019301 0.79970365      0.89353809
## Lag 9984      0.53583999      0.54805966 0.52360320      0.55069643
## Lag 49920      0.14306842      0.15452104 0.13684719      0.14980211
## Lag 99840      0.02458147      0.02539753 0.03095689      0.03477891
## Lag 499200      0.01843641      0.03279752 0.01720076      0.02073645
##      nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0      0.68130419      0.66364488      1.00000000
## Lag 9984      0.39279666      0.38433813      0.55548560
## Lag 49920      0.10152476      0.10370044      0.13805841
## Lag 99840      0.01755951      0.01172247      0.02069113
## Lag 499200      0.02488716      0.03174813      0.00880925
##      nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0      0.964974506      0.690756728      0.159164972
## Lag 9984      0.552596704      0.359103814      0.019720003
## Lag 49920      0.130705362      0.040193403      -0.037341598
## Lag 99840      0.021536248      -0.029337810      -0.030822237

```



```

## Lag 499200      0.005900843      -0.000598164      0.006140737
##      absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0      0.75116885      0.62241900      0.7451698453
## Lag 9984      0.40682865      0.30131374      0.3641937856
## Lag 49920      0.11040033      0.06553513      0.0820873896
## Lag 99840      0.01181873      0.02142308      -0.0005272461
## Lag 499200      0.03060235      -0.00304366      0.0037699312
##
## , , nodeicov.alcohol
##
##      edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0      0.93084898      0.87420557 0.76567224      0.87492251
## Lag 9984      0.51559533      0.52769673 0.50097193      0.53009089
## Lag 49920      0.13738686      0.14694569 0.13035861      0.14256536
## Lag 99840      0.02469241      0.02511116 0.02907798      0.03292231
## Lag 499200      0.01088637      0.02684134 0.01444047      0.01468181
##      nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0      0.67471626      0.64290097      0.964974506
## Lag 9984      0.37207461      0.36379485      0.544553387
## Lag 49920      0.09819158      0.09874627      0.135145343
## Lag 99840      0.02163391      0.01460760      0.019443110
## Lag 499200      0.01714031      0.02169631      0.004450136
##      nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0      1.0000000000      0.710987928      0.147607088
## Lag 9984      0.5415148671      0.352025528      0.013337837
## Lag 49920      0.1299407105      0.039227795      -0.038262100
## Lag 99840      0.0215461318      -0.025318024      -0.030164995
## Lag 499200      0.0004780972      -0.003673812      0.001903396
##      absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0      0.74125033      0.594710488      0.7309641615
## Lag 9984      0.39126418      0.284206025      0.3529623407
## Lag 49920      0.10443141      0.053619196      0.0776339996
## Lag 99840      0.01061755      0.021405538      0.0003354836
## Lag 499200      0.02033354      -0.008564635      -0.0068685982
##
## , , absdiff.alcohol
##
##      edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0      0.692827732      0.64481291 0.574603932      0.645812782
## Lag 9984      0.350870515      0.36161674 0.349126435      0.355967723
## Lag 49920      0.078609943      0.08368094 0.071127368      0.074077532
## Lag 99840      0.007595521      0.01366343 0.002309638      0.009130110
## Lag 499200      -0.015617394      0.00101576 -0.012933633      -0.007472874
##      nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0      0.50159226      0.477886799      0.69075673
## Lag 9984      0.25287641      0.245588040      0.35435773
## Lag 49920      0.05036675      0.051727092      0.07404586
## Lag 99840      0.02237220      0.016122797      -0.00147137
## Lag 499200      -0.01308470      -0.007239786      -0.02343212
##      nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0      0.710987928      1.00000000      0.1244836885
## Lag 9984      0.353174733      0.49129301      -0.0004842524
## Lag 49920      0.070450426      0.08524309      -0.0289717748
## Lag 99840      -0.004151531      -0.02098130      -0.0193067636

```

```

## Lag 499200      -0.027122439      -0.02222285      -0.0215989645
##               absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0           0.552774958           0.49967644           0.580015085
## Lag 9984        0.254853293           0.21987869           0.269088761
## Lag 49920       0.056570427           0.02470765           0.046659571
## Lag 99840       0.009996206           0.01601125          -0.007308401
## Lag 499200     -0.003546949          -0.03191089          -0.025230228
##
## , , edgescov.logdistance_matrix
##
##               edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0           0.16102644           0.14775809  0.134354114           0.13185077
## Lag 9984        0.02559120           0.03008345  0.037979096           0.02529893
## Lag 49920       -0.03524153          -0.02861590 -0.022727002          -0.03624145
## Lag 99840       -0.03416182          -0.02677941 -0.033286390          -0.02433027
## Lag 499200      0.01865764           0.02439434  0.007104083           0.01001660
##               nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0           0.127212927           0.12573955           0.15916497
## Lag 9984        0.014177629           0.01323248           0.03246140
## Lag 49920       -0.025901371          -0.02120741          -0.03151165
## Lag 99840       -0.009115348          -0.00589960          -0.04159276
## Lag 499200      0.014981544           0.01559950           0.02210914
##               nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0           0.14760709           0.124483688           1.00000000
## Lag 9984        0.02931811           0.023432431           0.45967970
## Lag 49920       -0.02928533          -0.005296082           0.13694186
## Lag 99840       -0.04170333          -0.012990613           0.01870156
## Lag 499200      0.03086929           0.010201718          -0.01642943
##               absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0           0.15713273           0.2413241842           0.237086205
## Lag 9984        0.04322411           0.0801623118           0.064512900
## Lag 49920       -0.02606158          -0.0002295858           0.019060113
## Lag 99840       -0.01624776          -0.0191922453          -0.004113588
## Lag 499200      0.04277479          -0.0052987449           0.005392024
##
## , , absdiff.age
##
##               edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0           0.80762722           0.75963499  0.675429949           0.759823636
## Lag 9984        0.43945722           0.45361814  0.430266887           0.448038052
## Lag 49920       0.11294056           0.11705310  0.108795009           0.115930653
## Lag 99840       0.01884759           0.01992253  0.033227767           0.024198886
## Lag 499200      0.01189943           0.02524715  0.006520559           0.008952702
##               nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0           0.65372540           0.63734963           0.7511688474
## Lag 9984        0.36424495           0.35500185           0.3986236864
## Lag 49920       0.09053847           0.09187619           0.0914908773
## Lag 99840       0.03280124           0.02594841           0.0076824572
## Lag 499200      0.02327899           0.02286909          -0.0005304064
##               nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0           0.741250331           0.552774958           0.157132728
## Lag 9984        0.390725622           0.260540202           0.036817654
## Lag 49920       0.083518768           0.029656835          -0.021348203
## Lag 99840       0.005232304          -0.031213409          -0.014618855

```

```

## Lag 499200      -0.004090412      -0.005558925      -0.007332552
##               absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0           1.000000000      0.560313904      0.622846851
## Lag 9984        0.509638180      0.268512799      0.287885826
## Lag 49920       0.120608742      0.076099624      0.074394287
## Lag 99840       0.007472767      0.027592382      0.012679301
## Lag 499200     0.025945118      -0.004193805      -0.005743277
##
## , , gwideg.fixed.0.6
##
##               edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0           0.680387792      0.637559992      0.63966316      0.613019361
## Lag 9984        0.351405604      0.370253077      0.41854021      0.380854789
## Lag 49920       0.092906538      0.098498573      0.09685123      0.099795196
## Lag 99840      -0.001593899      0.001682246      0.01358983      0.011200355
## Lag 499200    -0.007205787      0.004911611     -0.01843377     -0.009913117
##               nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0           0.510070246      0.492971555      0.622418997
## Lag 9984        0.272938408      0.263302062      0.310578936
## Lag 49920       0.065845188      0.069743832      0.068741314
## Lag 99840       0.007772329      0.004290772     -0.008412361
## Lag 499200     -0.005172011     -0.006012674     -0.011377894
##               nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0           0.594710488      0.49967644      0.241324184
## Lag 9984        0.302989453      0.23783533      0.066301820
## Lag 49920       0.063369553      0.04066205      0.003305711
## Lag 99840      -0.009409941     -0.03602500     -0.035064400
## Lag 499200     -0.012320130     -0.01676496     -0.015853393
##               absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0           0.560313904      1.000000000      6.824208e-01
## Lag 9984        0.283038937      0.335084184      3.258118e-01
## Lag 49920       0.073295833      0.076100273      9.589212e-02
## Lag 99840      -0.012033944      0.004097441     -1.888931e-06
## Lag 499200     0.002948913      -0.031443288     -2.752340e-02
##
## , , gwodeg.fixed.0.6
##
##               edges nodematch.gender      mutual gwesp.OTP.fixed.0.2
## Lag 0           0.76059023      0.713055328      0.679928842      0.695684130
## Lag 9984        0.38993384      0.410001532      0.437959209      0.417301038
## Lag 49920       0.10052674      0.106031386      0.100168534      0.105455927
## Lag 99840      -0.01301570     -0.016599450     -0.004041433     -0.002893334
## Lag 499200    -0.01633553     -0.002053631     -0.037162828     -0.020280746
##               nodeofactor.gender.1 nodeifactor.gender.1 nodeocov.alcohol
## Lag 0           0.558760998      0.5431416551      0.74516985
## Lag 9984        0.288755517      0.2790959840      0.37684610
## Lag 49920       0.062049759      0.0627607727      0.08929552
## Lag 99840      -0.008247813     -0.0176652611     -0.01300658
## Lag 499200     -0.009626422     -0.0003355269     -0.02137651
##               nodeicov.alcohol absdiff.alcohol edgescov.logdistance_matrix
## Lag 0           0.73096416      0.58001509      0.237086205
## Lag 9984        0.37134332      0.27399442      0.061072701
## Lag 49920       0.08035102      0.04700308      -0.003612299
## Lag 99840      -0.01405110     -0.04931586      -0.041430489

```

```
## Lag 499200      -0.02223063      -0.02978659      0.009745479
##               absdiff.age gwideg.fixed.0.6 gwodeg.fixed.0.6
## Lag 0          0.622846851      0.68242077      1.00000000
## Lag 9984       0.307995105      0.32345773      0.34530782
## Lag 49920     0.077534693      0.06006822      0.08781313
## Lag 99840     -0.013875070     -0.01030810     -0.01374157
## Lag 499200    0.003493861      -0.03914326     -0.02417090
```

From the lags we see that the model explores the state space very well, since autocorrelation becomes very close to 0.

(4)

```
model_2_gof <- gof(
  model_2,
  GOF = ~ idegree + odegree + espartners + distance
)
```

```
##
## Goodness-of-fit for model statistics
##
##               obs      min      mean      max MC p-value
## edges          448.0000 364.0000 439.8700 541.0000      0.76
## nodematch.gender 394.0000 312.0000 387.2200 501.0000      0.80
## mutual          124.0000  97.0000 122.3300 148.0000      0.98
## gwesp.OTP.fixed.0.2 338.8721 231.2946 328.6883 464.6392      0.76
## nodeofactor.gender.1 266.0000 202.0000 258.5900 339.0000      0.76
## nodeifactor.gender.1 266.0000 204.0000 257.9000 329.0000      0.78
## nodeocov.alcohol 1084.0000 871.0000 1063.2200 1326.0000      0.74
## nodeicov.alcohol 1136.0000 915.0000 1115.3100 1409.0000      0.84
## absdiff.alcohol  446.0000 360.0000 439.0500 534.0000      0.84
## edgescov.logdistance_matrix 97.1500 63.3600 100.2948 156.2300      0.86
## absdiff.age      146.0000 111.1000 144.7950 172.8000      0.88
## gwideg.fixed.0.6  193.1131 181.3664 192.7876 202.8506      0.84
## gwodeg.fixed.0.6  201.5419 189.2079 201.1498 210.0394      0.90
##
## Goodness-of-fit for minimum geodesic distance
##
##      obs  min   mean  max MC p-value
## 1    448  364  439.87  541      0.76
## 2    777  714 1097.00 2585      0.06
## 3   1162 1259 2112.92 3502      0.00
## 4   1580 1868 3060.26 4160      0.00
## 5   1802 2178 3169.11 3825      0.00
## 6   1739 1210 2340.58 3117      0.04
## 7   1625  375 1339.32 2284      0.54
## 8   1274  115  655.36 1324      0.06
## 9    872   10  292.48  858      0.00
## 10   573   0  121.74  648      0.02
## 11   287   0   47.77  369      0.04
## 12   103   0   16.75  198      0.08
```

```

## 13      21      0      5.87    97      0.16
## 14       5      0      1.97    54      0.18
## 15       0      0      0.56    21      1.00
## 16       0      0      0.16     9      1.00
## Inf 4244  758 1810.28 3698      0.00
##
## Goodness-of-fit for out-degree
##
##      obs min  mean max MC p-value
## 0       6   0  2.12   6      0.02
## 1       7   7 13.72  23      0.04
## 2      25  17 30.38  42      0.42
## 3      28  19 30.95  49      0.62
## 4      25  12 22.82  34      0.70
## 5      21   5 12.91  21      0.02
## 6      17   2  7.32  14      0.00
## 7       0   0  4.01  10      0.10
## 8       0   0  2.30   7      0.16
## 9       0   0  1.05   4      0.68
## 10      0   0  0.61   3      1.00
## 11      0   0  0.30   2      1.00
## 12      0   0  0.21   2      1.00
## 13      0   0  0.07   1      1.00
## 14      0   0  0.06   1      1.00
## 15      0   0  0.06   1      1.00
## 16      0   0  0.02   1      1.00
## 17      0   0  0.02   1      1.00
## 18      0   0  0.02   1      1.00
## 20      0   0  0.03   1      1.00
## 26      0   0  0.01   1      1.00
## 42      0   0  0.01   1      1.00
##
## Goodness-of-fit for in-degree
##
##      obs min  mean max MC p-value
## 0       6   2  6.70  14      1.00
## 1      18   8 16.67  26      0.90
## 2      28  13 26.52  47      0.74
## 3      23  16 25.95  40      0.66
## 4      18  11 20.62  29      0.56
## 5      12   6 13.01  23      0.94
## 6       9   3  8.38  20      0.96
## 7       8   0  4.65   9      0.10
## 8       2   0  2.71   7      1.00
## 9       4   0  1.48   5      0.16
## 10      0   0  0.93   5      0.80
## 11      1   0  0.57   3      0.86
## 12      0   0  0.31   2      1.00
## 13      0   0  0.23   2      1.00
## 14      0   0  0.09   1      1.00
## 15      0   0  0.05   1      1.00
## 16      0   0  0.02   1      1.00
## 17      0   0  0.05   1      1.00
## 18      0   0  0.03   1      1.00

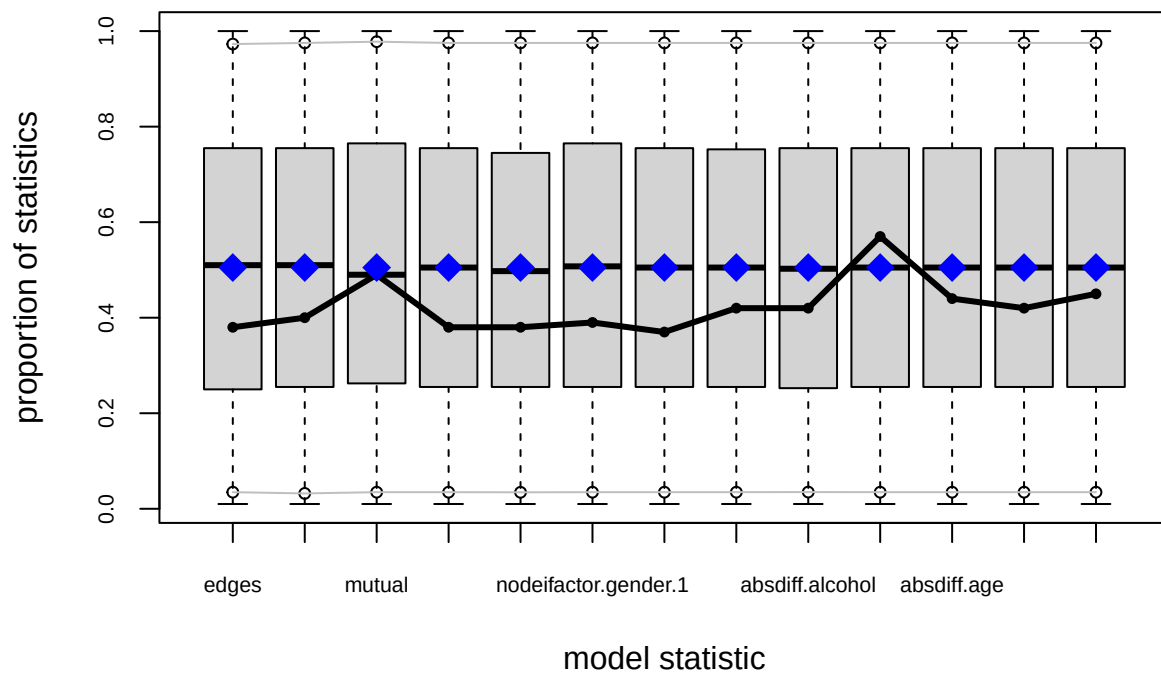
```

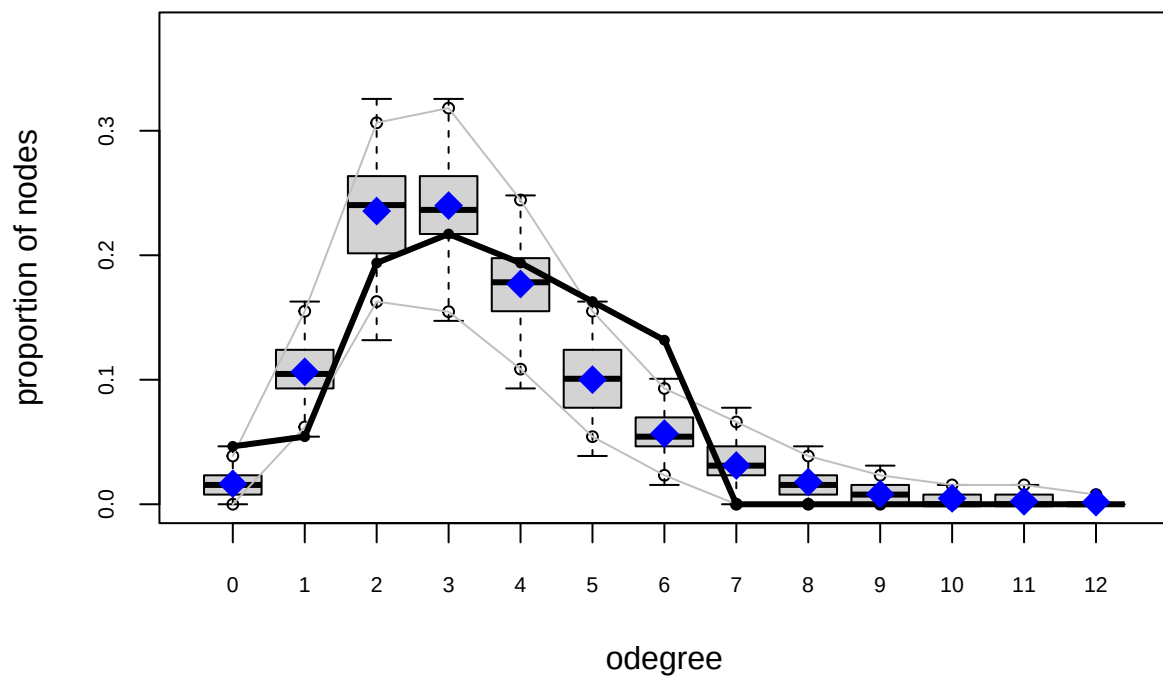
```

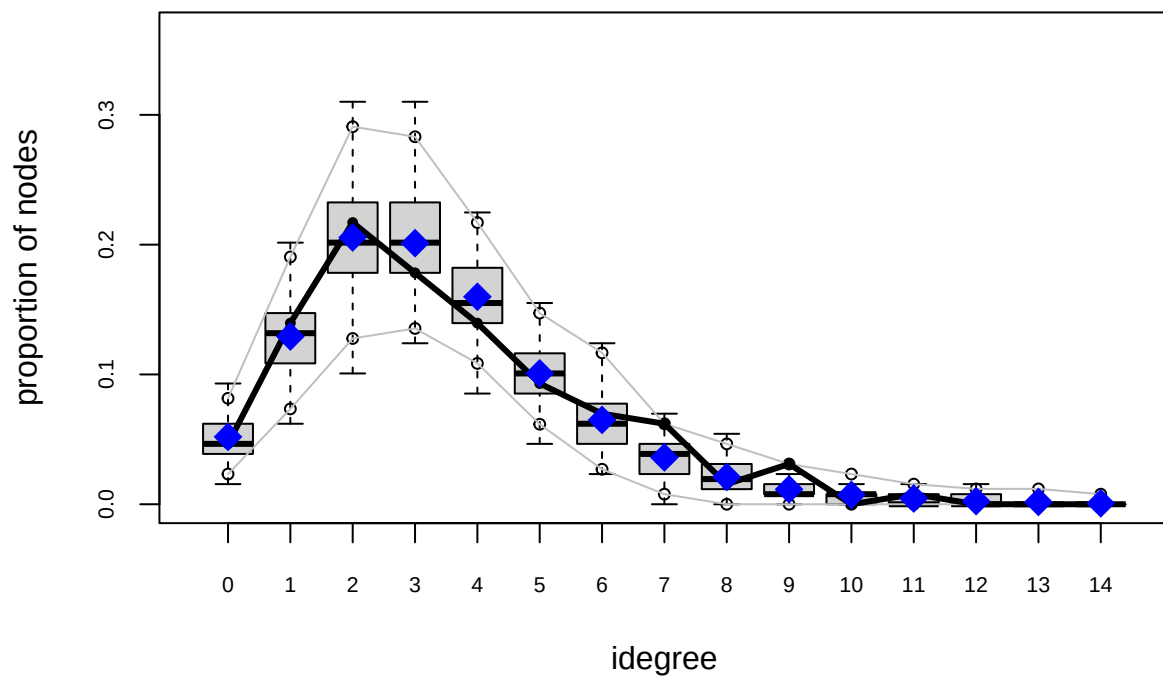
## 19  0  0  0.01  1      1.00
## 33  0  0  0.01  1      1.00
## 43  0  0  0.01  1      1.00
##
## Goodness-of-fit for edgewise shared partner
##
##      obs min   mean max MC p-value
## .0TP0 137  96 126.13 152    0.48
## .0TP1 167 164 233.72 285    0.02
## .0TP2  92  30  66.88 121    0.06
## .0TP3  42   1  11.30  37    0.00
## .0TP4   9   0   1.47  13    0.02
## .0TP5   1   0   0.30   5    0.36
## .0TP6   0   0   0.06   3    1.00
## .0TP8   0   0   0.01   1    1.00

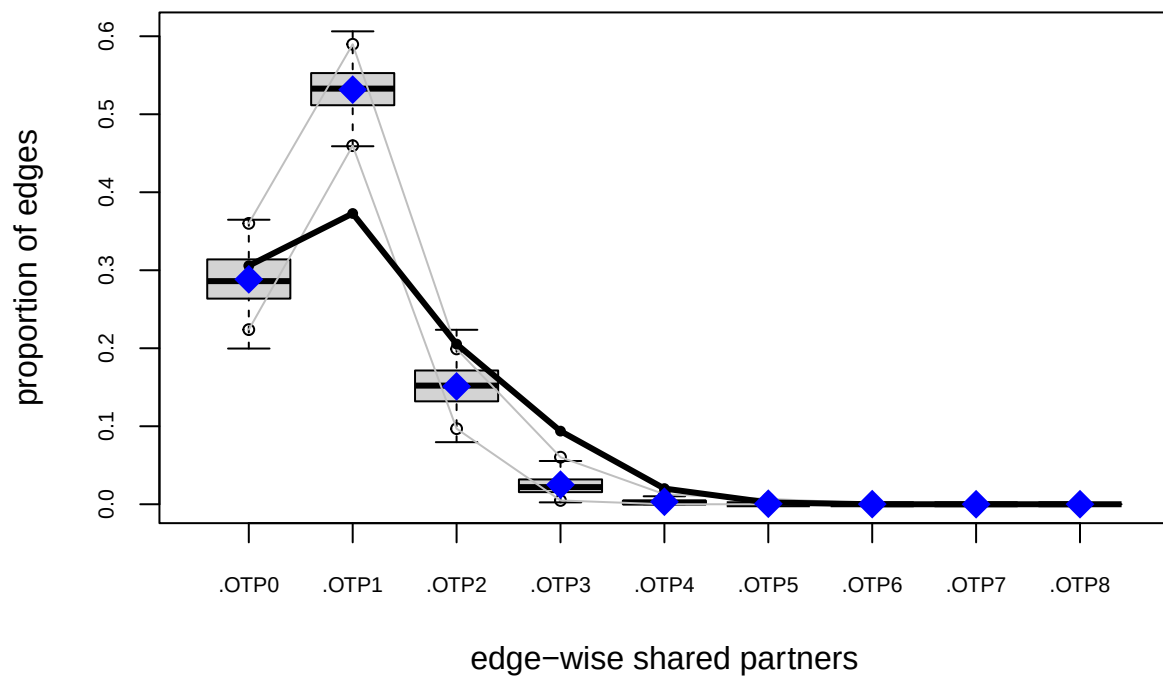
```

From the summary table we see that the p-values are high (not all of them are significant on the 5%) level. This also indicates that the model fits well based on the statistics that it is trained on.

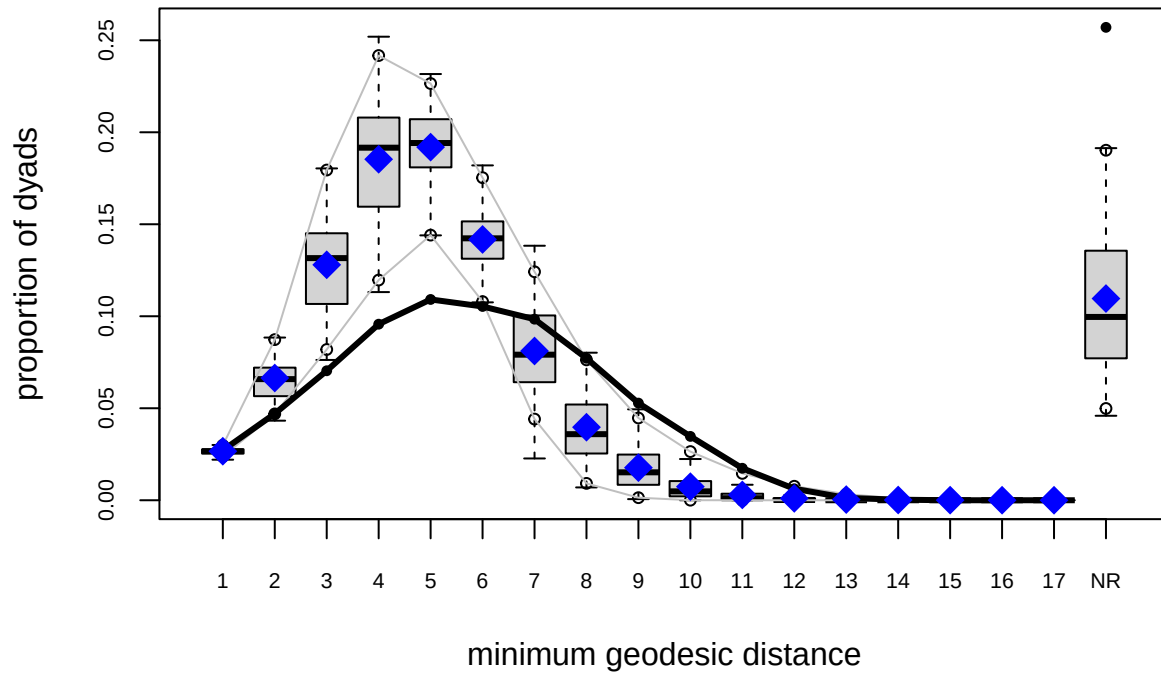








Goodness-of-fit diagnostics



Plotting the auxiliary statistics shows us that the model explores networks with similar in- and out-degree distributions, but the explored networks are much less connected than the reference one. This can be seen in the `distance` and `espartners` diagrams.

(5)

```
## Call:
## ergm(formula = ergm_model_2_params, control = control.ergm(MCMC.samplesize = 5000,
##   MCMC.interval = 10 * 1000, MCMC.burnin = 100 * 1000, init = rough_estimates,
##   parallel = 8, parallel.type = "PSOCK"))
##
## Monte Carlo Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -6.15915    0.20984    0 -29.352 < 1e-04 ***
## nodematch.gender    0.83015    0.10229    0  8.115 < 1e-04 ***
## mutual           2.80021    0.18541    0 15.103 < 1e-04 ***
## gwesp.OTP.fixed.0.2  1.71037    0.08780    0 19.481 < 1e-04 ***
## nodeofactor.gender.1 -0.07067    0.16074    0 -0.440 0.660173
## nodeifactor.gender.1  0.02701    0.14995    0  0.180 0.857035
## nodeocov.alcohol    -0.13034    0.06238    0 -2.089 0.036667 *
## nodeicov.alcohol     0.15214    0.05480    0  2.776 0.005496 **
## absdiff.alcohol     -0.12397    0.03900    0 -3.179 0.001477 **
## edgecov.logdistance_matrix -0.19896    0.05445    0 -3.654 0.000258 ***
## absdiff.age         -0.16849    0.14967    0 -1.126 0.260268
```

```
## gwideg.fixed.0.6          1.14908    0.31926      0   3.599 0.000319 ***
## gwodeg.fixed.0.6          2.52501    0.43956      0   5.744 < 1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
## Residual Deviance:  2779  on 16499  degrees of freedom
##
## AIC: 2805  BIC: 2905  (Smaller is better. MC Std. Err. = 0.5191)
```

- i - reciprocity is supported by data
- ii - transitivity (*gwesp.OTP.fixed.0.2*) is supported by the data
- iii, iv - cannot tell as the coefficients are insignificant
- v - the opposite is supported
- vi - supported
- vii - supported (negative coefficients, so the abs difference should be small)

(6)

???

(7)

- **H1** - Children with similar literacy scores tend to become friends (literacy score homophily)
- **H2** - Children with higher delinquency tend to send more friend nominations
- **H3 (a bit unsure about this one)** - Alcohol homophily is more prevalent in boys compared to girls